

MAULANA AZAD NATIONAL INSTITUTE OF TECHNOLOGY BHOPAL
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
End Term Examination, November 2025

Course: B. Tech.

Semester: Vth

Branch: EE

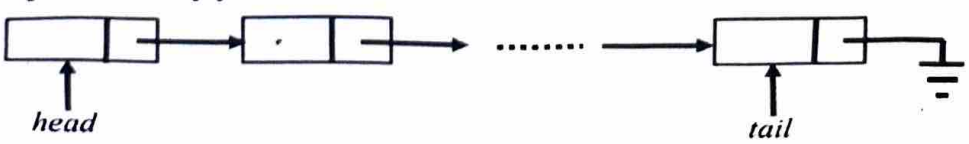
Subject: Data Structures and Algorithm

Subject Code: CS-352

Date: 17/11/2025

Time : 3 hr

Max Marks: 50

Q. No.	Question	Marks
1(A)	Following is the code of a binary search to find a key x in an array of size n. The initial array is in descending order. Some code is missing. You need to find the missing codes marked as A, B, C, D. Also, find the worst case time complexity of this code. <pre> int binarySearch_descending(int arr[], int n, int x) { int low = 0; int high = n-1; while (——A——) { int mid = ——B——; if (arr[mid] == x) return mid; if (arr[mid] < x) ——C—— else ——D—— } return -1; } </pre>	5
1(B)	Given an array, $A[10:19, -2:1, 5:10]$ with a base address of 1000 and the size of each element is 4 bytes in memory. Find the address of element $A[15][-1][9]$ when the array elements are stored in row-major order and column-major order.	5
2(A)	Convert this infix to postfix using stack. $P+(Q*R-(S/T^U)*V)*W$	5
2(B)	A queue is implemented using a non-circular singly linked list. The queue has a head pointer and a tail pointer, as shown in the figure. Let n denote the number of nodes in the queue. Let enqueue be implemented by inserting a new node at the head, and dequeue be implemented by deletion of a node from the tail. What will be the time complexity of this implementation of enqueue and dequeue. Justify your answer. 	5
3(A)	The following C function takes a single-linked list of integers as a parameter and rearranges the elements of the list. The function is called with the list containing	5

	the integers 1,2,3,4,5,6,7 in the given order. What will be the contents of the list after the function completes execution? <pre> struct node { int value; struct node *next; }; void rearrange (struct node *list){ struct node *p, * q; int temp; if(!list !list-> next) return; p = list; q = list->next; while (q) { temp = p->value; p-> value = q ->value; q-> value = temp; p = q-> next; q = p ? p->next : 0; } } </pre>	
3(B)	Insert the elements 17,9,34,56,11,71,86,55,22 in a hash table of size 9 using a hash function $h(k) = k \bmod 9$. Use linear probing to resolve a hashing collision and show step by step outcome.	5
4(A)	The pre-order traversal sequence of a binary search tree (BST) is 30, 20, 10, 15, 25, 23, 39, 35, 42. Find the post-order traversal sequence of the BST.	5
4(B)	<pre> graph TD 4((4)) --- 2((2)) 4 --- 7((7)) 2 --- 1((1)) 2 --- 3((3)) 7 --- 6((6)) 7 --- 17((17)) 6 --- 5((5)) 17 --- 14((14)) 17 --- 18((18)) </pre> Consider this initial AVL tree. Perform the insertion operation (Insert 8, 9) first and then delete operation (delete 7) on AVL tree. Show the final AVL tree.	5
5(A)	<pre> graph TD f((f)) --- b((b)) f --- d((d)) b --- d b --- a((a)) d --- a a --- c((c)) a --- e((e)) c --- g((g)) g --- e </pre> Find the DFS traversal sequence using a stack starting from node a. Show step by step, the visited node sequence and stack status.	5
5(B)	Sort this the array [5632,170, 435, 705, 90, 802, 24, 21, 66,5] in descending order using radix sort technique.	5